

**UBC Centre of Comparative Medicine (CCM)  
Plant Safety Relief Valve Try Lever Testing  
Procedure****Procedure #13****Prepared by:**  
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Clive Baek**Revised Date:** March 2026**Original Date:** October 2018**CCM PLANT SAFETY RELIEF VALVE TRY LEVER TESTING**

This procedure details the steps required to safely test the relief valves for the three Separate Steam Distribution systems and for the Cleaver Brooks and Clayton Boiler.

**This testing must be completed every 4 months on a schedule.**

| Safety Valve              | Safety Valve Setting | Safety Valve System Pressure | Safety Valve Serial Number | Initial when Complete | Date and Time when Complete |
|---------------------------|----------------------|------------------------------|----------------------------|-----------------------|-----------------------------|
| Clayton Boiler (B1) North | 150 PSI              | 120 PSI                      |                            |                       |                             |
| Clayton Boiler (B1) South | 150 PSI              | 120 PSI                      |                            |                       |                             |
| Cleaver Brooks (B2)       | 150 PSI              | 100 PSI                      |                            |                       |                             |
| 12# Steam System          | 15 PSI               | 12 PSI                       |                            |                       |                             |
| 45# Steam System          | 50 PSI               | 45 PSI                       |                            |                       |                             |
| 80# Steam System          | 90 PSI               | 80 PSI                       |                            |                       |                             |

The purpose of this procedure is to confirm relief valve internals are free to move and function effectively as necessary to assure the safety and security of personnel, systems and equipment. This test must be conducted with the line pressure within 75% of the safety valve setting to ensure a proper test and to allow the valve to reseal properly.

Testing of these valves provides a level of assurance that they are available and will operate as necessary to protect the assigned systems. Failure of these valves to function effectively as required can lead to equipment failure with the potential for injury to personnel.

**UNDER NO CIRCUMSTANCES MUST ANY ADJUSTMENTS BE MADE TO ANY VALVE SETTINGS TO ASSIST/FORCE IT TO CLOSE.**

When testing or working around safety valves exercise care and attention to assure that valve seals are not broken. If a valve seal is broken it must be reported to the supervisor having responsibility for the area. This person has the responsibility for ensuring that through the line of authority the Boilers and Pressure Vessels

**Personnel in the area must be informed that the test will be taking place and what to expect.**

| Step | Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Initial when Complete | Date and Time when Complete |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|
| 1    | <p>Check and assure that the safety relief valve drain pan drain is clear. For this check pour a litre of water into the drain pan and check the outlet.</p> <p><b>NOTE:</b> If the drain is not continuous, drape the local area around the safety relief valve drain with a see through curtain able to withstand temperatures to 300 degrees Fahrenheit for increase personnel safety</p>                                                                                                                                        |                       |                             |
| 2    | Check that there is at least a ¼" gap between the drain pan and the vent line.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                       |                             |
| 3    | The person performing the test signals to the other person a readiness to perform the test.                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |                             |
| 4    | <p>Test the valve by pulling the valve-lifting lever in a direction such that the lifting lever end describes a short arc. The lift shall be short and sharp with a quick release of tension on the lever. This to result in a three to ten second blow.</p> <p>The valve may not lift due to mechanical malfunction or to conditions arising from the environment (dust rings, corrosion, etc.). Attempt one more lift. If the valve does not function, immediately stop the test and report the situation to your Supervisor.</p> |                       |                             |
| 5    | <p>After 'blow' monitor valve reseating. The valve will simmer for a few seconds after the blow before reseating properly. This condition is due to valve disc distortion during the blow.</p> <p>If the valve does not reseat after test, repeat steps 3 and 4 to try and reseat this valve. If the valve still doesn't reseat, shutoff the boiler or corresponding line and advise your supervisor.</p>                                                                                                                           |                       |                             |
| 6    | <p>The safety relief valve test is now complete.</p> <p>Log the date and time of the test in CCM Plant logbook</p>                                                                                                                                                                                                                                                                                                                                                                                                                  |                       |                             |
| 7    | This is the end of the procedure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                       |                             |